

Yashwanth Karthik

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EDUCATION

PES University, EC Campus

B.Tech in Computer Science and Engineering

Bengaluru, India

Aug. 2025 – Present

NPS Whitefield

Higher Secondary (12th) and Secondary (10th), CBSE

Bengaluru, India

Completed

EXPERIENCE

Team Vegavath, PES University

Robotics Vice Head

Bengaluru, India

Oct. 2025 – Present

- Leading club robotics engineering initiatives and 15kg featherweight combat robot development
- Contributed to revamp of the Vegavath club website using Next.js, TypeScript, and Supabase

PESUIO Course – Quantum Computing with Qiskit

Course Participant

PES University

Oct. 2025 – Nov. 2025

- Completed hands-on coursework on quantum computing using IBM's Qiskit Python toolkit
- Explored quantum gates, circuit construction, superposition, and entanglement algorithms

PROJECTS

Tinarchy | *Python, Linux, Nginx, Server-Sent Events, Tailscale, systemd*

Aug. 2026 – Present

- Architected a modular dashboard and server control center with real-time SSE delta telemetry streaming (<500B dynamic frames, >50% bandwidth reduction) and thread-safe in-memory API micro-caching (up to 134x speedup)
- Engineered direct Tailscale UNIX domain socket resolution (<2ms WHOIS latency), autonomous closed-loop thermal PID governor, multicore network packet steering (RPS/RFS), and adaptive ZRAM/disk-swap kernel autotuning

SpectreFlow | *Python, Docker, Radare2, YARA, NetworkX, Tkinter, pefile* | *GENESYS 2.0*

Mar. 2026

- Built an end-to-end malware analysis suite combining static binary inspection, dynamic process/network/filesystem monitoring, and containerized Docker sandboxing for the GENESYS 2.0 hackathon at PES University
- Implemented PE section entropy analysis, YARA threat scanning, Control Flow Graph (CFG) generation via Radare2, automated risk scoring, and VirusTotal / MalwareBazaar threat intelligence lookup

mat.nvim | *Rust, Neovim Lua API, pulldown-cmark, syntect*

Sep. 2026

- Engineered a sub-5ms in-editor Markdown preview plugin for Neovim powered by a standalone Rust rendering engine with TrueColor ANSI syntax highlighting, zero browser overhead, and zero runtime dependencies
- Implemented single-keystroke in-place mode switching between rendered view and raw editing buffer with cursor line sync, live unwritten buffer previewing, and full GFM table/callout parsing

niri-rice | *Rust, Wayland, Niri, Shell, CSS, Linux*

Jul. 2026 – Present

- Built a modular, power-optimized scrollable tiling Wayland desktop environment featuring dynamic Matugen Material You palette generation across UI components, terminals, and lockscreens
- Developed `rice-ctl` in Rust for sub-second system theming, wallpaper transitions, and automated multi-package installation scripts with custom Waybar and PAM fingerprint unlock

QState-Arena | *Python, Tkinter, Qiskit, Matplotlib, NumPy*

Nov. 2025

- Built a gamified 2-qubit quantum state simulator with interactive 3D Bloch sphere visualization, entanglement glow effects, and 5 progressive puzzle levels from $|00\rangle$ to Bell states
- Integrated Qiskit AerSimulator for real-time statevector tracking, fidelity checking, and 200-shot measurement simulation

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, Bash/Zsh, HTML/CSS

Frameworks & Libraries: React, Next.js, Tailwind CSS, Tokio, Clap, Tkinter, Flask, Django, OkHttp, Qiskit

Developer Tools: Git, GitHub Actions, Linux, Neovim, Docker, Android Studio, VS Code, Playwright, Nginx

Core Domains: Systems Programming, Event-Driven Architecture, Full-Stack Development, Security Analysis